

(1-2) Molecular Geometry (VSEPR)

- Lewis structure → count central atom's e groups = bonded atoms + lone pairs (double/triple bond or radical e = 1 group)

e ⁻	Hyb (e geo)	B	LP	Molecular geo	Angle	Pol
2	sp (linear)	2	0	linear	180°	x
3	sp ² (tri. planar)	3	0	trigonal planar	120°	x
3	sp ²	2	1	bent	<120°	✓
4	sp ³ (tetrahedral)	4	0	tetrahedral	109.5°	x
4	sp ³	3	1	trigonal pyramidal	<109.5°	✓
4	sp ³	2	2	bent	<109.5°	✓
5	sp ³ d (tri. bipy.)	5	0	trigonal bipyramidal	90/120°	x
5	sp ³ d	4	1	seesaw	90/120°	✓
5	sp ³ d	3	2	T-shaped	90, 180°	✓
5	sp ³ d	2	3	linear	180°	x
6	sp ³ d ² (octahedral)	6	0	octahedral	90°	x
6	sp ³ d ²	5	1	square pyramidal	90°	✓
6	sp ³ d ²	4	2	square planar	90°	x

- Lone pairs squeeze angles (SCI a little < 109.5°) · SO₃, BF₃ tri. planar · PCl₃, PH₃, NH₃ tri. pyr. · CS₂, CO₂, N₂, I₂ linear · SiH₄, CH₄, PO₄³⁻ tetra. · SCI₂, H₂S, H₂O, O₃, ClNO, NH₂ bent · ClF₃ T-shaped · XeF₄ sq. planar

(3-4) Polarity & Dipole Moment

- Dipole** = separation of + / - charge; net dipole moment ($\mu \neq 0$) = **polar** molecule
- Polar = polar bonds (ΔEN) whose vectors **don't cancel**. Pol column above assumes identical outer atoms — **mixed outer atoms** → **polar** (CH₃Cl, CH₂Cl₂, CHCl₃)
- Nonpolar: same-element diatomic (Br₂), all hydrocarbons, x-row shapes: XeF₄, XeF₆, CCl₄, PBr₃, PCl₅, BeCl₂, CO₂, BF₃, BH₃, SO₃
- Polar: different-element diatomic (CO, IBr, HCl) + lone-pair shapes: BrF₃ (sq. pyr.), NH₃, OF₂, H₂O, SO₂ (bent), NCl₃, NF₃
- $\uparrow \Delta EN = \uparrow \mu$: CH₃F > CH₃Cl > CH₃Br > CH₃I · NH₃ > PH₃ > AsH₃ (EN falls down group)
- Polar dissolves polar: NH₃ in H₂O ✓, H₂S in NH₃ ✓, CCl₄/SiO₂/O₂/CO₂ in polar x

(5-6) Hybridization of an Atom in a Structure

- Count e groups **on that atom** (bonded atoms + lone pairs) → table (1): 2 sp, 3 sp², 4 sp³, 5 sp³d, 6 sp³d²
- sp² geometry is trigonal planar (**not** square planar)
- C, 4 single bonds → **sp³** · C with 1 double bond (C=O, C=C, benzene ring) → **sp²** · C with triple or 2 doubles → **sp**
- O in -O-H or ether (2 atoms + 2 LP) → **sp³** · carbonyl =O → **sp²**
- N with 3 atoms + 1 LP (amine) → **sp³**
- Each atom separately: CH₃-CHO → left C sp³, carbonyl C sp²

(7-8) σ and π Bonds: Counting & Concepts

- single = 1 σ · double = 1 σ + 1 π · triple = 1 σ + 2 π**
- Count **every** bond, incl. each C-H! σ = # of atom-atom connections; π = extras on doubles/triples
- CH₃-CH=CH-COOH: 11 σ , 2 π · benzene C₆H₆: 12 σ (6 C-C + 6 C-H) + 3 π
- σ : end-to-end overlap, e density **on** internuclear axis, rotation free, stronger
- π : **side-by-side p-p** overlap, density above & below axis → **restricted rotation** (energy needed to break the p-p interaction) → cis/trans in C=C (1,2-dichloroethene), none around C-C single (1,2-dichloroethane)
- Two 1s (or head-on p+p along bond axis) → σ and σ^* · sideways p+p (perpendicular axis) → π and π^*
- Benzene delocalized π MOs: e **free to move around the ring**; all C-C equal

(9-10) Intermolecular Forces

Force	Who has it
Dispersion (London, van der Waals, induced dipole)	ALL molecules; $\uparrow$ w/ size & molar mass; straight-chain > branched
Dipole-dipole	polar molecules
H-bond	H bonded directly to N, O, F
Ion-dipole	ion + polar solvent (NaCl in H ₂ O)

- Dispersion only (nonpolar): Br₂, Cl₂, F₂, H₂, CO₂, BCl₃, CF₄, hydrocarbons, noble gases
- Disp + dip-dip: HCl, HBr, SO₂, CHCl₃, CH₃Br, ethers (CH₃OCH₃), CH₃CHO, CH₃CN
- + H-bond: H₂O, NH₃, HF, CH₃OH, CH₃CH₂OH, CH₃NH₂, CH₃COOH, H₂O₂, HOCH₂CH₂OH
- Traps: H-bond **with water** just needs N/O/F (CH₃COOH ✓, CH₃NH₂ ✓; C₂H₆ x; Na⁺ = ion-dipole) · S never H-bonds (H₂S, CH₃SCH₃) · ether O has no H → none with itself · polar **bonds** ≠ polar molecule (CO₂)

(11-12) Rank Vapor Pressure & Boiling Point

- Strength: **ionic > H-bond > dipole-dipole > dispersion**, but size can trump: big nonpolar can beat small polar
- $\uparrow$ IMF → $\uparrow$ bp, $\uparrow$ mp, $\uparrow$ viscosity, $\uparrow$ surf. tension, $\uparrow$ ΔH_{vap} , $\uparrow$ critical T, $\downarrow$ **vapor pressure**
- VP order = reverse of bp order** (lowest bp = highest VP at room T)
- Weak IMF ⇒ low ΔH_{vap} , high VP, low bp

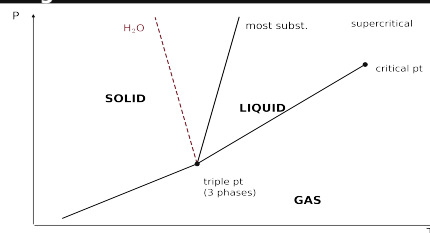
VP $\downarrow$	CF ₄ > PF ₃ > BrF ₃ (polar last)
bp $\uparrow$	CH ₄ < CH ₃ Cl < CH ₃ OH < RbCl (ionic)
bp (disp. size)	I ₂ > Br ₂ > Cl ₂ > F ₂ · CH ₃ CHO > CH ₃ OCH ₃ > Rn
bp (H-bond)	HF > HI > HBr > HCl
bp (# of OH)	HOCH ₂ CH ₂ OH > ClCH ₂ CH ₂ OH > CH ₃ CH ₂ OH > CH ₃ CH ₂ Cl

(13) Viscosity, Capillary Action, Surface Tension

Surface tension	energy required to stretch/increase surface area by unit amount
Viscosity	resistance of a liquid to flow ; $\uparrow$ with IMF, $\downarrow$ with $\uparrow$ T; spherical/compact → lower
Capillary action	liquid flows up a thin tube against gravity = cohesion (like molecules) + adhesion (liquid-glass)

- Concave meniscus (H₂O): adhesion > cohesion; convex (Hg): cohesion > adhesion. Highest viscosity = most H-bonding (HOCH₂CH₂OH)

(14) Phase Diagrams



- Lines = 2 phases coexist · **triple point** = all 3 · **critical point**: above T_c no pressure can liquefy gas → **supercritical fluid**
- $\uparrow$ T at const P (→): solid → liquid → gas. $\uparrow$ P at const T ($\uparrow$): gas → liquid → solid. Normal bp/mp read at 1 atm
- H₂O**: fusion line leans **left** (ice less dense) → $\uparrow$ P melts ice · **CO₂**: triple pt at 5.1 atm → at 1 atm solid **sublimes** s→g (dry ice)

(15-16) Type of Solid

Type	Held by	Properties	Examples
Ionic	± attraction	hard, brittle, high mp, poor cond.	NaCl, AgCl, CaF ₂ , MgO, CaCO ₃
Molecular	IMFs	soft, low mp, poor cond.	ice H ₂ O, CO ₂ , I ₂ , sucrose
Metallic (atomic)	e sea	low-high mp, good cond.	Au, Cu, Fe, Na
Network covalent (atomic)	covalent bonds	v. hard, v. high mp, poor cond.	diamond, quartz SiO ₂ , graphite, SiC
Nonbonding (atomic)	dispersion	v. low mp	noble gases: Ar(s)
Amorphous	no long-range order	melts over a range	glass, rubber, plastic, butter, wax
Ceramic	inorganic nonmetal; powder + H ₂ O + heat	brittle, heat-resistant	clay/kaolinite, porcelain, Al ₂ O ₃

- Graphite sp² sheets: conducts along sheets, slippery · diamond sp³: insulator · fcc packing = most space-efficient (12 neighbors)

(17) Band Theory & Doping

- Bands = MOs delocalized over the whole crystal: **valence band** = bonding MOs, **conduction band** = antibonding MOs, **band gap** between
- Conductor** (metal): bands continuous/overlap · **Semiconductor** (metalloids Si, Ge): small gap, e promotable · **Insulator**: **LARGE gap**
- Doping** = add impurity to $\uparrow$ conductivity:

	Dopant	Carrier
n-type	Group V/5A (P, As)	extra e ⁻ in conduction band (– charges)
p-type	Group III/3A (B, Al, Ga, In)	holes in valence band (e ⁻ hop between holes)
p-n junction	–	diode : current flows one way

(18) Heat of Solution & Hydration

$$\Delta H_{\text{solution}} = \Delta H_{\text{hydration}} - \Delta H_{\text{lattice energy}}$$

- $\Delta H_{\text{solution}}$** : heat absorbed/released on dissolving = $\Delta H(\text{solvent-solvent}) + \Delta H(\text{solute-solute}) + \Delta H(\text{solvent-solute})$
- $\Delta H_{\text{hydration}}$** : heat **released** when 1 mol **gaseous ions** dissolves in water (forms ion-dipole; always exo). $\Delta H_{\text{solute}} = -\Delta H_{\text{lattice}}$
- Highest ΔH_{hydr}** : smallest ion + highest charge: Al³⁺ > Mg²⁺ > Na⁺ > K⁺ · F⁻ > Cl⁻
- ΔH_{soln}** **exo** (–): $\Delta H_{\text{hydration}} > \Delta H_{\text{solute}}$ · **endo** (+): solute > hydration · ≈ 0 : comparable

(19) Saturation; Like Dissolves Like

Unsaturated	less than equilibrium amount — more can dissolve
Saturated	max amount at that T; dissolved $\rightleftharpoons$ solid dynamic equilibrium
Supersaturated	more than equilibrium amount; unstable — seed crystal/scratch $\rightarrow$ excess crystallizes out

- Solution forms if solute-solvent attraction **comparable** to solute-solute + solvent-solvent · polar/ionic $\leftrightarrow$ polar (H_2O , ethanol, NH_3); nonpolar $\leftrightarrow$ nonpolar (hexane, CCl_4 , benzene)
- Most soluble in ethanol = most H-bonds (ethylene glycol); in hexane = longest chain (1-pentanol); all **nitrates** soluble

(20) Solubility vs P & T; When Bubbles/Boiling

$\uparrow$ gives	Gas	Solid / liquid
$\uparrow$ Pressure	$\uparrow$ solubility (Henry)	no effect
$\uparrow$ Temperature	$\downarrow$ solubility	$\uparrow$ solubility

- Dissolve the most gas: **low T + high P** (cold soda holds CO_2)
- Liquid **boils** when its vapor pressure = **external (atm) pressure**; gas **bubbles out** of solution when its partial (vapor) pressure **> atm pressure**

(21) Molarity (and Molality)

$$M = \frac{\text{mol solute}}{L \text{ solution}} \quad m = \frac{\text{mol solute}}{\text{kg solvent}}$$

- mol = $M \times V(L) \cdot V(L) = \text{mol} \div M \cdot \text{mL} \div 1000 = L \cdot g \leftrightarrow \text{mol}$ via molar mass · dilution: $M_1 V_1 = M_2 V_2$
- 0.250 L of 0.500 M $\rightarrow$ 0.125 mol $\cdot$ need 0.300 mol of 0.450 M $\rightarrow$ 0.667 L = 667 mL
- Molality m uses **kg of solvent**, not solution — for $\Delta T_f / \Delta T_b$. Only molarity changes with T (volume swells); m, mass %, χ don't

(22) Percent Concentration (% w/v)

$$\% w/v = \frac{g \text{ solute}}{100 \text{ mL soln}} \Rightarrow g = \frac{\% \times \text{mL}}{100}$$

- 1.5% (w/v), 250 mL $\rightarrow 1.5 \times 250/100 = \mathbf{3.75 \text{ g}}$ (n.b. the summary's "31.25 g" contradicts its own formula — trust $g = \% \times \text{mL} / 100$)
- mL from % w/v: 16.6 g $\div$ 0.0450 g/mL = 369 mL
- % m/m** = g solute / (g solute + g solvent) $\times$ 100 · g water = g soln $\times$ (1 - %/100): 300 g of 9.71% KBr $\rightarrow$ 271 g water
- % v/v** = vol component / total vol $\times$ 100

(23) Henry's Law ($C_{\text{gas}} = k_H P_{\text{gas}}$)

$$C_{\text{gas}} = k_H P_{\text{gas}} \quad k_H = C_{\text{gas}} / P_{\text{gas}}$$

- k_H = Henry's constant = "solubility constant of a gas" (M/atm); C = molar solubility
- mmHg $\rightarrow$ atm: $\div 760$
- $k_H = C/P$: 1.3 M at 0.022 atm $\rightarrow$ **59 M/atm**
- $4.7\text{e-}4$ M at 522 mmHg $\rightarrow k_H = 4.7\text{e-}4 \div (522/760) = \mathbf{6.8\text{e-}4 \text{ M/atm}}$

(24) Osmosis

- Osmosis**: flow of **solvent** (water) across a semipermeable membrane from **low solute conc.** $\rightarrow$ **high solute conc.** (dilute $\rightarrow$ concentrated)
- Semipermeable membrane**: passes solvent, **blocks solute** (ions) — it does *not* stop water
- Hyperosmotic** = more solute: water flows **into** it (cell in it shrinks/crenates)
- Hyposmotic** = less solute: water flows **out** into the stronger side (cell in it swells/bursts)
- Isosmotic** = equal: no net flow (isotonic saline = 0.9% w/v NaCl) · n = pressure needed to **stop** osmotic flow

(25) WORD: MO Theory — Bond Order & Magnetism

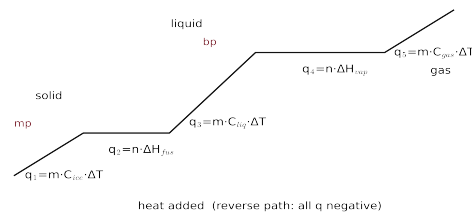
B_2	C_2	N_2	O_2 filled (F_2 , Ne_2 , NO , OF , Cl_2 ...)
σ_{2p}^+	_____	_____	σ_{2p}^+
π_{2p}^+	_____	_____	π_{2p}^+
σ_{2p}	_____	_____	π_{2p}
π_{2p}	_____	_____	σ_{2p}
σ_{2s}^+	_____	_____	σ_{2s}^+
σ_{2s}	_____	_____	σ_{2s}

- Valence e⁻ total: **add 1 per - charge, remove 1 per +**
- Fill low $\rightarrow$ high, 2 per orbital, Hund across the n pair
- BO = (bonding e⁻ - antibonding e⁻) / 2**
- Any unpaired e⁻ $\rightarrow$ **paramagnetic**; all paired $\rightarrow$ **diamagnetic**
- BO > 0 $\rightarrow$ stable, can form (H_2 1 $\checkmark$; He_2 0 $\times$ can't exist); **higher BO = stronger & shorter** bond (N_2 weaker/longer than N_2). σ_{2s} + σ_{2s}^* cancel — you may count 2p only

Sp.	e ⁻	BO	Mag	Sp.	e ⁻	BO	Mag
N_2	10	3	dia	O_2	12	2	para (2 $\uparrow$)
N_2^+	9	2.5	para	O_2^+	11	2.5	para
N_2^-	11	2.5	para	O_2^-	13	1.5	para
F_2, Cl_2	14	1	dia	Cl_2^+	13	1.5	para
B_2	6	1	para	Cl_2^-	15	0.5	para

- Heteronuclear (NO , OF) $\rightarrow$ the O-type (right) ordering (more EN atom sits lower) · period 3 (Cl_2 , Na_2) $\rightarrow$ same shapes, n = 3

(26) WORD: Phase-Change Heat (gas $\rightleftharpoons$ solid)



- Sketch the path; skip segments you don't cross (65 °C liq $\rightarrow$ 115 °C gas: heat liq 65 $\rightarrow$ 100, vaporize, heat gas 100 $\rightarrow$ 115)
- Sloped (one phase): **q = m(g)·Cs·ΔT** — flat (phase change, T constant): **q = n(mol)·ΔH**
- Sum; convert J $\rightarrow$ kJ. Cooling path (gas $\rightarrow$ solid): every q **negative** ($\Delta H_{\text{cond}} = -\Delta H_{\text{vap}}$, $\Delta H_{\text{freeze}} = -\Delta H_{\text{fus}}$)
 - H_2O : C(ice) 2.09, C(liq) 4.184, C(steam) $\approx$ 2.0 J/g·°C; ΔH_{fus} 6.02, ΔH_{vap} 40.7 kJ/mol; MM 18.02
 - 36.0 g H_2O , 65 °C liq $\rightarrow$ 115 °C gas: $36 \times 4.184 \times 35 = 5.27 \text{ kJ}$; $(36/18.02) \times 40.7 = 81.3 \text{ kJ}$; $36 \times 2.01 \times 15 = 1.09 \text{ kJ} \rightarrow \mathbf{87.7 \text{ kJ}}$
 - Cooling: 125 g benzene (78.11) gas 425 K $\rightarrow$ liq 335 K (bp 353 K, mp 279 K — never freezes): $125 \times 1.06 \times 72 = 9.5 \text{ kJ}$; $(125/78.11) \times 33.9 = 54.3 \text{ kJ}$; $125 \times 1.73 \times 18 = 3.9 \text{ kJ} \rightarrow \mathbf{67.7 \text{ kJ removed}}$
 - Endo (+): melt, vaporize, sublime · Exo (-): freeze, condense, deposit · $\Delta H_{\text{sub}} = \Delta H_{\text{fus}} + \Delta H_{\text{vap}}$

(27) WORD: Raoult's Law (VP Lowering)

$$P_{\text{soln}} = \chi_{\text{solvent}} P_{\text{solvent}}^\circ \quad \Delta P = \chi_{\text{solute}} P_{\text{solvent}}^\circ$$

$$\chi_{\text{solvent}} = \frac{\text{mol}_{\text{solvent}}}{\text{mol}_{\text{solvent}} + i \cdot \text{mol}_{\text{solute}}}$$

- mol solvent = g $\div$ MM; mol solute = g $\div$ MM (**x i if ionic**)
- $\chi(\text{solute}) = \text{mol solvent} \div \text{total mol}$ — always < 1
- $P(\text{soln}) = \chi(\text{solvent}) \times P^\circ$; lowering $\Delta P = P^\circ - P(\text{soln}) = \chi(\text{solute}) \cdot P^\circ$
 - 75.0 g citric acid (192.12) + 420 g H_2O , P° 71.93 mmHg: 0.390 & 23.31 mol $\rightarrow \chi = 0.9835 \rightarrow \mathbf{70.7 \text{ mmHg}}$
 - 200 g sucrose (342.3) + 350 g H_2O , P° 17.5: 0.584 & 19.42 mol $\rightarrow \chi = 0.9708 \rightarrow \mathbf{17.0 \text{ mmHg}}$
 - Both volatile: $P(\text{total}) = \chi_A \cdot P^\circ_A + \chi_B \cdot P^\circ_B$

(28) WORD: Freezing $\downarrow$ / Boiling $\uparrow$ (van't Hoff)

$$\Delta T_f = i K_f m \quad \Delta T_b = i K_b m$$

- m = mol solute / **kg solvent** · H_2O : K_f 1.86, K_b 0.512 °C/m · $T_f = T^\circ_f - \Delta T_f$, $T_b = T^\circ_b + \Delta T_b$
- i** = particles per formula unit: nonelectrolytes (glucose, sucrose, urea) 1 · NaCl, KCl, NaOH, NH_4Cl 2 · CaCl_2 , $(\text{NH}_4)_2\text{SO}_4$ 3 · $\text{Al}(\text{NO}_3)_3$, FeCl_3 4 (measured i slightly less — ion pairing)
- Find i: **$i = \Delta T_b / (K_b \cdot m)$** — 1.83 m boils 102.5 °C: $2.5 / (0.512 \times 1.83) = \mathbf{2.7}$
- Molar mass from fp: $\Delta T_f \rightarrow m = \Delta T_f / (i \cdot K_f) \rightarrow \text{mol} = m \times \text{kg} \rightarrow \mathbf{MM = g/mol}$: 20.0 g per 500 g benzene (T°_f 5.444, K_f 5.12) freezes 3.77 °C: ΔT 1.674 $\rightarrow$ 0.327 m $\rightarrow$ 0.163 mol $\rightarrow \mathbf{\approx 120 \text{ g/mol}}$
- Lowest fp, equal grams = most mol $\times$ i (small MM wins: NaOH > KCl, CaCl_2)

(29) WORD: Osmotic Pressure $\rightarrow$ Molar Mass

$$\pi = i M R T \Rightarrow M = \frac{\pi}{i R T}$$

- R = **0.08206** L·atm/mol·K · T in **K** (+273.15) · n in **atm** (torr $\div$ 760) · i = 1 nonelectrolyte
- 1. $M = n / (iRT)$
- 2. mol = $M \times V(L)$
- 3. **MM = g solute $\div$ mol**
 - 15.2 **mg** in 150.0 mL, n 8.44 torr, 25 °C: $M = (8.44/760) / (0.08206 \times 298) = 4.54\text{e-}4 \rightarrow 6.81\text{e-}5 \text{ mol} \rightarrow 0.0152/6.81\text{e-}5 = \mathbf{223 \text{ g/mol}}$
 - n $\rightarrow$ freezing pt: M from n $\rightarrow$ per 1 L: g solute = $M \times \text{MM}$; g soln = 1000 $\times$ density; kg solvent = (g soln - g solute)/1000 $\rightarrow$ molality $\rightarrow \Delta T_f = i \cdot K_f \cdot m$